Existence of a noncircular oval with convex unit-tangent iterates

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Allowed standard prerequisites (editorial).

- Classical geometric research prerequisites cited by the author. Complete original embedded bibliography retained. Standard prerequisites remain cited; all source-local carrying estimates, constructions and non-adjacent proof continuations are retained.

1 Introduction

The unit-tangent transform sends a circle of radius r to a circle of radius $\sqrt{r^2+1}$. We show that a noncircular oval can also remain convex under iteration.

Let γ be a smooth regular closed plane curve parametrized by arclength, and let $\tau = \gamma'$ be its unit tangent. The *unit-tangent transform* is the oriented curve

$$\mathcal{T}\gamma = \gamma + \tau.$$

Curves are identified under orientation-preserving regular reparametrization, so $\mathcal{T}$ is independent of parametrization, and we reparametrize by arclength at each step. Convex curves are oriented counterclockwise, with inward unit normal ν and curvature k defined by $\tau' = k\nu$. A convex closed curve is the boundary of a convex body, traversed once. An *oval* is a smooth convex closed curve with $k > 0$ everywhere.

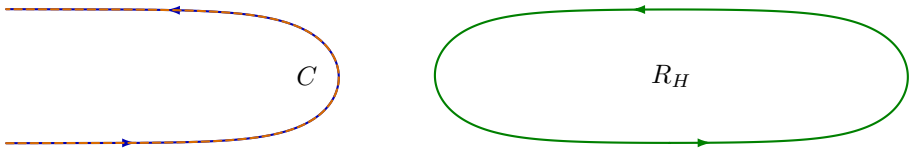
Tabachnikov asked whether an oval all of whose unit-tangent iterates are convex must be a circle [Tab15, Section 9], [Tab23, Conjecture 4].

Theorem 1.1. *There is a noncircular oval Γ_0 such that $\mathcal{T}^n\Gamma_0$ is an oval for every integer $n \geq 0$.*

The construction uses long ovals whose width in one direction stays bounded. Their local geometry comes from a convex hairpin that $\mathcal{T}$ carries to a horizontal translate (see Figure 1).

In bicycle kinematics, the curves R and $F = \mathcal{T}R$ are the rear and front tracks of a unit segment. The steering equation describes the angle between their tangents [LT09, Section 2] (see also [BLPT20, Proposition 2.6]). For a convex closed front F of curvature less than 1, this equation selects a unique convex rear $\mathcal{B}F$ on the branch where the steering angle lies between 0 and $\pi/2$. We establish the continuity and regularity of the inverse map $\mathcal{B}$ in Section 2.

Section 3 constructs the translating hairpin, and Section 4 periodizes it to obtain closed pairs $\mathcal{T}R_H = F_H$ with front half-perimeter H and rear half-perimeter $P(H) = H - \Delta + o(1)$ for a constant $\Delta > 0$. Section 5 shows that the curvature functions of R_H and $F_{P(H)}$, expressed in suitably chosen arclength coordinates, are exponentially close in L^1. Section 6 turns these comparisons into an exact orbit, and Section 7 uses the uniform width bound to exclude circles.



(a) (b)

Figure 1: (a) A translating hairpin C (solid) and its unit-tangent image $\mathcal{T}C$ (dashed). (b) A closed rear curve R_H.

2 The inverse on convex curves

For an angle ψ, write $\tau(\psi) = (\cos\psi, \sin\psi)$. Suppose that $F = \mathcal{T}R$. Write s for front arclength and x for rear arclength, let Θ and Ψ be the corresponding tangent angles, and let $\delta = \Theta - \Psi$ be the steering angle. Resolving the front velocity in the rear Frenet frame gives

$$x_s = \cos\delta, \qquad \Psi_s = \sin\delta.$$

If K and k are the front and rear curvatures, then

$$K = \delta_s + \sin\delta, \qquad k = \tan\delta.$$

Curves converge in geometric C^r if they have orientation-preserving parametrizations, defined on a fixed circle, that converge in C^r. The inverse $\mathcal{B}$ remains defined even when the front curvature vanishes at some points.

Lemma 2.1. *Let F be a regular convex closed C^2 curve with curvature $0 \le K \le K_0 < 1$. There is a unique periodic solution of*

$$\delta_s = K - \sin\delta, \qquad 0 \le \delta \le \arcsin K_0.$$

It is positive and defines a regular convex closed rear $\mathcal{B}F$ with curvature

$$0 < k \le \frac{K_0}{\sqrt{1 - K_0^2}}.$$

The map $\mathcal{B}$ preserves central symmetry and its center, is continuous in geometric C^2, and carries smooth families of fronts to smooth families of rears. Every C^r front with $r \ge 2$ has a C^{r+1} rear.

Proof. The period map advances a solution by one circuit of length L. Since the steering vector field points into the interval $I = [0, \arcsin K_0]$ at both endpoints, this map sends I into itself. By the mean value theorem, the difference w of two solutions in I satisfies

$$w_s = -a(s)w, \qquad a(s) \ge \sqrt{1 - K_0^2} > 0,$$

so the period map is a contraction on I and therefore has a unique fixed point. Every periodic solution on the larger branch $0 < \delta < \pi/2$ also lies in I, because at a maximum of δ the differential equation gives $\sin\delta \le K_0$.

Since $\delta_s + \delta = K + \delta - \sin\delta \ge K$, an integrating factor over a period gives

$$(1 - e^{-L})\delta(s) \ge \int_{s-L}^{s} e^{t-s} K(t)\, dt > 0.$$

The last inequality is strict because $K \geq 0$ and $\int_0^L K(s)\,ds = 2\pi$.

Choose a tangent-angle lift Θ for F, put $\Psi = \Theta - \delta$, and define

$$R(s) = F(s) - \tau(\Psi(s)).$$

Then $R_s = \cos\delta\,\tau(\Psi)$, so R is regular and closed, and $\mathcal{T}R = F$. Since $0 < \delta \leq \arcsin K_0$, the curvature $k = \tan\delta$ of R satisfies $0 < k \leq K_0/\sqrt{1-K_0^2}$. Because δ is periodic, the tangent of R turns through 2π, so R is embedded and bounds a strictly convex body. By uniqueness, if F is centrally symmetric, then δ has period $L/2$ and $\tau(\Psi)$ reverses after $L/2$, so R is centrally symmetric with the same center.

In normalized front arclength, the period map for fronts near F is a uniform contraction that varies continuously with K. Thus its fixed point varies continuously with K, and the periodic solution δ depends continuously in C^1 on the front in C^2. Since $x_s = \cos\delta > 0$, the rear and its arclength parametrization also depend continuously on the front in C^2. The same argument gives smooth dependence on parameters.

For regularity, a C^r front has a C^{r-1} tangent angle Θ. The equation $\Psi_s = \sin(\Theta - \Psi)$ gives a C^r solution Ψ, and $x_s = \cos(\Theta - \Psi) > 0$ gives a C^r change of arclength. Thus the rear tangent is C^r in rear arclength and the rear curve is C^{r+1}. □

3 A translating hairpin

In this section we construct a hairpin C such that $\mathcal{T}C = C + (V, 0)$. Here a *hairpin* is an embedded curve of positive curvature, complete in arclength in both directions, whose tangent turns through π and whose ends are asymptotic to parallel lines.

Parametrize C by its tangent angle $\theta \in (0, \pi)$, let $\rho(\theta)$ be its radius of curvature, and set

$$C_\theta = \rho(\theta)\tau(\theta), \qquad f(\theta) = \rho(\theta)\sin\theta.$$

Writing $C = (X, Z)$ gives

$$X' = f\cot\theta, \qquad Z' = f. \tag{3.1}$$

Since $(C + \tau)' = (f/\sin\theta)\tau + \nu$, the tangent angle of $C + \tau$ is $g(\theta) = \theta + d(\theta)$, where $d(\theta) = \arctan(\sin\theta/f(\theta))$.

Lemma 3.1. *Let $f \in C^\infty(0,\pi)$ satisfy $f \geq m > 1$. If*

$$\int_\theta^{\theta + d(\theta)} f(t)\,dt = \sin\theta, \qquad d(\theta) = \arctan\frac{\sin\theta}{f(\theta)}, \tag{3.2}$$

then $g = \theta + d$ is an increasing diffeomorphism of $(0,\pi)$, and the curve C defined by (3.1) satisfies

$$C(\theta) + \tau(\theta) = C(g(\theta)) + (V, 0) \tag{3.3}$$

for a constant V.

Proof. Differentiating (3.2) gives

$$f(g)\,g' = f + \cos\theta > 0.$$

Since $0 < d \leq \sin\theta/m < \pi - \theta$, the endpoint limits of g are 0 and π, so g is an increasing diffeomorphism of $(0,\pi)$. The integral identity gives the vertical component of (3.3).

For the horizontal component, the cotangent addition formula with $\cot d = f/\sin\theta$ gives $(f + \cos\theta)\cot g = f\cot\theta - \sin\theta$. Since $(X \circ g)' = f(g)g'\cot g = (f + \cos\theta)\cot g$, we obtain

$$\frac{d}{d\theta}\big(X(\theta) + \cos\theta - X(g(\theta))\big) = f\cot\theta - \sin\theta - (f + \cos\theta)\cot g = 0.$$ □

For a bounded measurable function $f : (0,\pi) \to \mathbf{R}$ with $f \geq m > 1$, set $A_f(t) = \int_0^t f(u)\,du$. Since $A_f(\pi) - A_f(\theta) \geq m(\pi - \theta) > \sin\theta$ for $0 < \theta < \pi$, the function

$$D_f(\theta) = A_f^{-1}(A_f(\theta) + \sin\theta) - \theta$$

is well defined on $(0,\pi)$, and we set

$$(\mathcal{P}f)(\theta) = \sin\theta\,\cot D_f(\theta).$$

Lemma 3.2. *For every bounded measurable function $f : (0,\pi) \to \mathbf{R}$ with $f \geq m > 1$, the function $\mathcal{P}f$ is positive and continuous on $(0,\pi)$. Moreover, $\mathcal{P}$ is order preserving, and $f = \mathcal{P}f$ if and only if f satisfies* (3.2).

Proof. The primitive A_f and its inverse are continuous, and $0 < D_f \leq \sin\theta/m < \pi/2$, so $\mathcal{P}f$ is positive and continuous. If $f \leq h$, then the integral of h reaches $\sin\theta$ no later than that of f, so $D_h \leq D_f$ and $\mathcal{P}f \leq \mathcal{P}h$. Finally, $f = \mathcal{P}f$ is equivalent to (3.2) by the definition of D_f. □

To construct a fixed point, define the barrier functions

$$f_\varepsilon^{\pm}(\theta) = \varepsilon^{-1} + \frac{2}{3}(1 - \cos\theta) \pm \varepsilon(1 + \cos\theta).$$

For small $\varepsilon > 0$, both are bounded below by $\varepsilon^{-1} - 2\varepsilon > 1$, and $f_\varepsilon^+ - f_\varepsilon^- = 2\varepsilon(1 + \cos\theta) \geq 0$.

Lemma 3.3. *For all sufficiently small $\varepsilon > 0$,*

$$f_\varepsilon^- \leq \mathcal{P}f_\varepsilon^- \leq \mathcal{P}f_\varepsilon^+ \leq f_\varepsilon^+ \qquad (0 < \theta < \pi).$$

Proof. For $d = \arctan(\sin\theta/f(\theta))$, put

$$\mathcal{R}(f) = \int_\theta^{\theta+d} f(t)\,dt - \sin\theta.$$

The integral increases with its upper limit, and cotangent decreases. Hence

$$\mathcal{R}(f) \geq 0 \iff D_f \leq d \iff \mathcal{P}f \geq f,$$

and the reversed inequalities are equivalent as well.

Each barrier has the form $f(\theta) = A + B\cos\theta$. Exact integration gives

$$\mathcal{R}(f) = f(\theta)\,d - \sin\theta + B\sin\theta\,(\cos d - 1) + B\cos\theta\,(\sin d - d).$$

Since $f(\theta) = \sin\theta\,\cot d$,

$$\frac{2\mathcal{R}(f)}{d^2\sin\theta} = \frac{2(d\cot d - 1)}{d^2} + B\frac{2(\cos d - 1)}{d^2} + \frac{B\cos\theta}{f(\theta)}\frac{2(\sin d - d)}{d^2\tan d}.$$

In the variable d, the three quotients extend to even analytic functions at $d = 0$ with values $-2/3$, -1, and $-1/3$ and remainders $O(d^2)$. For $f = f_\varepsilon^{\pm}$ with $b = \pm 1$, we have $B = -2/3 + \varepsilon b$, $f(\theta)^{-1} = \varepsilon + O(\varepsilon^2)$, and $0 < d \leq 2\varepsilon\sin\theta$, so Taylor's theorem at $d = 0$ gives

$$\begin{aligned}\frac{2\mathcal{R}(f)}{d^2\sin\theta} &= -\frac{2}{3} - B - \frac{B\cos\theta}{3f(\theta)} + O(\varepsilon^2)\\ &= \varepsilon\left(\frac{2}{9}\cos\theta - b\right) + O(\varepsilon^2),\end{aligned}$$

uniformly for $0 < \theta < \pi$. Thus the leading coefficient is at least $7/9$ for f_ε^- ($b = -1$), which gives $f_\varepsilon^- \leq \mathcal{P}f_\varepsilon^-$, and at most $-7/9$ for f_ε^+ ($b = 1$), which gives $\mathcal{P}f_\varepsilon^+ \leq f_\varepsilon^+$. Lemma 3.2 gives $\mathcal{P}f_\varepsilon^- \leq \mathcal{P}f_\varepsilon^+$. □

Theorem 3.4. *For every sufficiently small $\varepsilon > 0$, equation (3.2) has a solution $f \in C^\infty(0,\pi)$ with $f_\varepsilon^- \le f \le f_\varepsilon^+$. The corresponding curve C is a hairpin in a strip of finite width and satisfies*

$$\mathcal{T}C = C + (V, 0)$$

for some $V > 0$.

Proof. Starting with $f_0 = f_\varepsilon^-$, define $f_{n+1} = \mathcal{P} f_n$. Lemmas 3.2 and 3.3 give

$$f_\varepsilon^- \le f_0 \le f_1 \le \cdots \le f_\varepsilon^+.$$

Let $f = \lim_n f_n$ on $(0,\pi)$. The functions f_n share a lower bound $m > 1$, and monotone convergence gives $\int_0^\pi (f - f_n)(t)\,dt \to 0$. Since $f_n \le f$, we have $D_{f_n} \ge D_f$, and comparing the two defining integrals yields

$$0 \le D_{f_n}(\theta) - D_f(\theta) \le \frac{1}{m} \int_0^\pi (f - f_n)(t)\,dt \longrightarrow 0.$$

Passing to the limit in $f_{n+1} = \mathcal{P} f_n$ gives $f = \mathcal{P} f$, so f satisfies (3.2).

Since the fixed-point identity makes f continuous, A_f and its inverse are C^1, and hence so are D_f and f. Iterating this argument gives $f \in C^\infty(0,\pi)$.

Define C by (3.1). Its radius of curvature is $f/\sin\theta > 0$, and $Z' > 0$ makes it embedded. Since $m \le f \le M$ for some $M < \infty$, at both ends the curve has infinite arclength, X tends to $-\infty$, and Z has a finite limit. Thus the ends are asymptotic to horizontal lines. Lemma 3.1 gives $\mathcal{T}C = C + (V, 0)$, and evaluating (3.3) at $\theta = \pi/2$ yields

$$V = -\int_{\pi/2}^{g(\pi/2)} f(t) \cot t\,dt > 0. \qquad \square$$

Fix one such hairpin with curvature less than $1/20$. All constants below may depend on this fixed choice and on the indicated derivative orders, but not on the period introduced in the next section.

Regard the hairpin as a rear track and its translate as the front. In front arclength s, let

$$y(s) = \sin\delta(s), \qquad c(s) = \cos\delta(s) = \sqrt{1 - y(s)^2}.$$

Choose arclength origins so that the hairpin and its translate have the same curvature function K_*. If $x = x(s)$ is the corresponding rear arclength, then

$$x' = c, \qquad K_*(s) = y + \frac{y'}{c}, \qquad K_*(x(s)) = \frac{y}{c}. \tag{3.4}$$

Lemma 3.5. *There is a constant $a > 0$ such that, for every integer $j \ge 0$, there are constants $C_j, D_j > 0$ satisfying*

$$|y^{(j)}(s)| \le C_j e^{-a|s|}, \qquad |y^{(j)}(s)| \le D_j\, y(s). \tag{3.5}$$

Moreover, $K_*(s) \ge b_0\, y(s)$ for a constant $b_0 > 0$, and

$$\int_{\mathbf{R}} y(s)\,ds = \pi, \qquad \Delta = \int_{\mathbf{R}} (1 - c(s))\,ds > 0.$$

Proof. Let u be rear arclength and $\theta(u)$ the rear tangent angle, and set $r(u) = \tan(\theta(u)/2)$. Then

$$(\log r)_u = \frac{\theta_u}{\sin\theta} = \frac{1}{f(\theta)}, \qquad K_*(u) = \frac{\sin\theta(u)}{f(\theta(u))}.$$

Since $m \le f \le M$, both $\sin\theta$ and K_* have two-sided exponential bounds at the ends. By (3.4), $0 < 1 - x' = 1 - c \le y^2/c = K_*(x)^2 x'$. Substituting $u = x(s)$ gives

$$0 < \Delta = \int_{\mathbf{R}} (1 - c(s))\, ds \le \int_{\mathbf{R}} K_*(u)^2\, du < \infty,$$

so $s - x(s)$ is bounded on $\mathbf{R}$. Because $(\log r)_u = 1/f$ is bounded, shifting the argument by $s - x(s)$ changes r, $\sin\theta = 2r/(1+r^2)$, and $K_* = \sin\theta/f$ by bounded factors. Thus

$$K_*(s) \ge b_0\, K_*(x(s)) = b_0 \frac{y(s)}{c(s)} \ge b_0\, y(s)$$

and $K_*(s) = O(y(s))$, together with the exponential bound for $y(s)$ in (3.5).

We prove $|y^{(j)}(s)| \le D_j\, y(s)$ by induction on $j \ge 0$. Take $D_0 = 1$, and suppose that the bound holds through order j. Since $0 < y < 1/20$ and $c = \sqrt{1 - y^2}$, through order j the derivatives of c are bounded and those of y/c are $O(y)$. Applying $(c^{-1}\partial_s)^i$ to $K_*(x(s)) = y(s)/c(s)$ yields $K_*^{(i)}(x(s)) = O(y(s)) = O(K_*(x(s)))$ for $0 \le i \le j$. Since $x(\mathbf{R}) = \mathbf{R}$, we have $K_*^{(i)} = O(K_*) = O(y)$ for $0 \le i \le j$. Taking the jth derivative of $y' = c\,(K_* - y)$ then gives $|y^{(j+1)}| \le D_{j+1}\, y$, completing the induction and proving (3.5). Finally, the change of variables $y(s)\, ds = K_*(u)\, du$ gives $\int_{\mathbf{R}} y(s)\, ds = \pi$, since the rear tangent turns through π. □

4 Closed rear–front pairs

For a period $H > 0$, periodize $y = \sin\delta$ by setting

$$Y_H(s) = \sum_{m \in \mathbf{Z}} y(s - mH).$$

For all sufficiently large H, we have $0 < Y_H < 1$. Put $\delta_H = \arcsin Y_H$ and $c_H = \sqrt{1 - Y_H^2}$, so that $\delta_H' = Y_H'/c_H$, and set

$$K_H = Y_H + \frac{Y_H'}{c_H} = Y_H + \delta_H', \qquad k_H(x_H(s)) = \frac{Y_H(s)}{c_H(s)}, \qquad x_H' = c_H.$$

Let

$$I_H = [-H/2, H/2], \qquad \overline{K}_H(s) = \sum_{m \in \mathbf{Z}} K_*(s - mH).$$

Lemma 4.1. *There is a constant $\beta > 0$ such that, for every integer $r \ge 0$, there is a constant $C_r > 0$ satisfying*

$$\|Y_H - y\|_{C^r(I_H)} \le C_r e^{-\beta H} \tag{4.1}$$

for all sufficiently large H.

Proof. For $s \in I_H$ and $m \ne 0$, we have $|s - mH| \ge (|m| - 1/2)H$, so the exponential bounds in Lemma 3.5 justify termwise differentiation and summation and give (4.1). □

Lemma 4.2. *For all sufficiently large H,*

$$|K_H(s) - \overline{K}_H(s)| \le C e^{-\beta H} Y_H(s).$$

Proof. Put $y_m(s) = y(s - mH)$ and $G(z) = (1 - z^2)^{-1/2}$. The formulas for K_H and K_* give

$$K_H - \overline{K}_H = \sum_{m \in \mathbf{Z}} y_m'(G(Y_H) - G(y_m)).$$

For large H, all arguments of G lie in a fixed interval $[0, b]$ with $b < 1$, on which G is Lipschitz. Since $|y_m'| \le D_1 y_m$ by (3.5), the sum is bounded in absolute value by $C \sum_{m \neq n} y_m\, y_n$. For $n = m + \ell$ with $\ell \neq 0$, at least one of $|s - mH|$ and $|s - nH|$ is at least $|\ell| H/2$, so (3.5) gives

$$y_m\, y_n \le C e^{-a|\ell|H/2}(y_m + y_n).$$

Summing over m and then over $\ell \neq 0$ completes the proof. □

Proposition 4.3. *For all sufficiently large H, the functions k_H and K_H are the curvatures of smooth centrally symmetric ovals R_H and F_H with $\mathcal{T} R_H = F_H$ and $R_H = \mathcal{B} F_H$. Their perimeters are $2P(H)$ and $2H$, respectively, where*

$$P(H) = \int_0^H c_H(s)\, ds = H - \Delta + O(e^{-\beta H}).$$

The function P is continuous, and the curvatures of both curves are at most $1/10$.

Proof. Choose Θ_H with $\Theta_H' = K_H$, and define F_H by $F_H' = \tau(\Theta_H)$. Since δ_H is H-periodic and $\int_0^H Y_H(s)\, ds = \int_{\mathbf{R}} y(s)\, ds = \pi$,

$$\int_0^H K_H\, ds = \int_0^H (\delta_H' + Y_H)\, ds = \pi.$$

The tangent reverses after each interval of length H, so F_H closes after two periods and is centrally symmetric. Set

$$\Psi_H = \Theta_H - \delta_H, \qquad R_H = F_H - \tau(\Psi_H).$$

Then $\Psi_H' = Y_H$ and $R_H' = c_H\, \tau(\Psi_H)$, so R_H is the corresponding rear, also closed and centrally symmetric, with half-perimeter $P(H)$ and positive curvature k_H. Since $\tau(\Psi_H)$ also reverses after each interval of length H, the curves R_H and F_H have the same center of symmetry.

Lemmas 3.5 and 4.2 give

$$K_H \ge \overline{K}_H - C e^{-\beta H} Y_H \ge (b_0 - C e^{-\beta H}) Y_H > 0$$

for large H. Both closed curves have positive curvature and turning number 1, and hence are ovals. On I_H, estimate (4.1) gives

$$K_H = K_* + O(e^{-\beta H}), \qquad \frac{Y_H}{c_H} = \frac{y}{c} + O(e^{-\beta H}).$$

Since the front and rear hairpin curvatures K_* and y/c are less than $1/20$, periodicity bounds the curvatures of R_H and F_H by $1/10$ for all sufficiently large H, and Lemma 2.1 gives $R_H = \mathcal{B} F_H$.

By (4.1), $c_H = c + O(e^{-\beta H})$ on I_H, whereas $1 - c = O(y^2)$ decays exponentially outside I_H. Since c_H is H-periodic, we have

$$H - P(H) = \int_{I_H} (1 - c_H(s))\, ds = \int_{\mathbf{R}} (1 - c(s))\, ds + O(e^{-\beta H}) = \Delta + O(e^{-\beta H}),$$

where decreasing β absorbs polynomial factors in H. The same estimates show that P is continuous. □

The width of a curve in a direction is the length of its orthogonal projection onto a line in that direction.

Lemma 4.4. *There is a constant $C_W > 0$ such that, for all sufficiently large H, the oval F_H has width at most C_W in some direction.*

Proof. Rotate F_H so that its tangent angle Θ_H increases from 0 to π on I_H. Its vertical width is

$$W_H = \int_{-H/2}^{H/2} \sin\Theta_H(s)\, ds.$$

On I_H, Lemmas 3.5 and 4.2 give $K_H(s) \le CY_H(s) \le C'e^{-a|s|}$, so

$$\sin\Theta_H(s) \le \min\{\Theta_H(s),\, \pi - \Theta_H(s)\} = \min\left\{\int_{-H/2}^{s} K_H(t)\, dt,\, \int_{s}^{H/2} K_H(t)\, dt\right\} \le \frac{C'}{a} e^{-a|s|}.$$

Integrating over I_H gives $W_H \le 2C'/a^2$. □

5 Matching curvature functions

By Proposition 4.3, the curves R_H and $F_{P(H)}$ have the same perimeter $2P(H)$. Keep the arclength origins chosen in (3.4), so that the hairpin and its translate have curvature function K_*, and normalize the closed rear arclength coordinate by $x_H(0) = x(0)$. We compare both k_H and $K_{P(H)}$ with the periodization $\overline{K}_{P(H)}$.

Theorem 5.1. *There are constants $C, \beta > 0$ such that, for all sufficiently large H,*

$$\int_{\mathbf{R}/P(H)\mathbf{Z}} |k_H(u) - K_{P(H)}(u)|\, du \le Ce^{-\beta H}.$$

Proof. Put $P = P(H)$ and $J_H = x_H(I_H)$. The interval J_H has length P, so its translates by $P\mathbf{Z}$ tile the line.

On I_H, Lemma 4.1 and the positive lower bounds for c_H and c give

$$\|c_H - c\|_{L^\infty(I_H)} + \|Y_H/c_H - y/c\|_{L^\infty(I_H)} \le Ce^{-\beta H}.$$

Integrating $x_H' - x' = c_H - c$ with $x_H(0) = x(0)$ yields $\|x_H - x\|_{L^\infty(I_H)} \le CHe^{-\beta H}$. Since $k_H(x_H) = Y_H/c_H$, $K_*(x) = y/c$, and K_*' is bounded,

$$\|k_H - K_*\|_{L^\infty(J_H)} = \|k_H(x_H) - K_*(x_H)\|_{L^\infty(I_H)} \le CHe^{-\beta H}.$$

Because $P \le H$ and $K_* > 0$, unfolding the tail of $\overline{K}_P$ gives

$$\|k_H - \overline{K}_P\|_{L^1(J_H)} \le P\|k_H - K_*\|_{L^\infty(J_H)} + \int_{\mathbf{R}\setminus J_H} K_*(u)\, du.$$

Since $x(s) - s$ and $x_H(s) - x(s)$ are bounded on I_H, the endpoints of J_H are $\pm H/2 + O(1)$, so the tail integral is exponentially small. Decreasing β to absorb polynomial factors in H gives

$$\|k_H - \overline{K}_P\|_{L^1(\mathbf{R}/P\mathbf{Z})} \le Ce^{-\beta H}.$$

For the front, Lemma 4.2 gives

$$|K_P - \overline{K}_P| \le Ce^{-\beta P} Y_P.$$

Integrating over a period and using $\int_0^P Y_P(s)\, ds = \pi$ yields

$$\|K_P - \overline{K}_P\|_{L^1(\mathbf{R}/P\mathbf{Z})} \le Ce^{-\beta P} \le C'e^{-\beta H}.$$

The triangle inequality completes the proof. □

6 Constructing the exact orbit

To construct an exact orbit near a sequence of *comparison curves* Q_n, we join each Q_n to $A_n = \mathcal{B}Q_{n+1}$ by an interpolation path and repeatedly apply $\mathcal{B}$ to these paths. Under the hypotheses of Theorem 6.4 below, the resulting normal variations are summable.

A centrally symmetric curve is *centered* if its center of symmetry is the origin. Throughout this section, curves are centered and oriented. Arclength origins are auxiliary choices used only to compare curvature functions, and changing an origin does not change the curve.

For a smooth path Γ of curves parametrized by $X(t,u)$, write

$$\partial_t X = \xi\tau + \eta\nu.$$

The normal velocity η is independent of the curve parameter. Define

$$W(\Gamma) = \int_0^1 \|\eta\|_{L^1}\,dt, \qquad S_j(\Gamma) = \int_0^1 \|\partial_s^j \eta\|_{L^\infty}\,dt \quad (j=0,1,2),$$

where s is arclength on the curve at time t and the norms are taken with respect to s. These quantities are additive under concatenation, and the same definitions apply to paths that are piecewise smooth in t.

Lemma 6.1. *Let G_0 and G_1 be centered ovals with the same half-perimeter L and curvature at most K_0. Choose arclength origins at which the tangent directions of G_0 and G_1 agree. Then there is a smooth path Γ of centered ovals from G_0 to G_1 with curvature at most K_0 such that*

$$W(\Gamma) + S_0(\Gamma) + S_1(\Gamma) \le C(1+L)^2 \|k_{G_1} - k_{G_0}\|_{L^1(0,L)}.$$

The constant C depends only on K_0.

Proof. Set $k_t = (1-t)k_{G_0} + tk_{G_1}$, and let

$$\theta_t(s) = \theta_0 + \int_0^s k_t(u)\,du,$$

where θ_0 is the common initial tangent angle. Define

$$X_t(s) = \int_0^s \tau(\theta_t(u))\,du - \frac{1}{2}\int_0^L \tau(\theta_t(u))\,du \qquad (0 \le s \le L),$$

and extend by $X_t(s+L) = -X_t(s)$. Each curvature function is L-periodic and has integral π, so this formula defines a smooth centered oval of curvature k_t, parametrized by arclength.

Put $\varepsilon = \|k_{G_1} - k_{G_0}\|_{L^1(0,L)}$. Then $\|\dot\theta_t\|_\infty \le \varepsilon$ and $\|\dot X_t\|_\infty \le (3/2)L\varepsilon$. For $\dot X_t = \xi\tau + \eta\nu$, we have $|\xi|, |\eta| \le |\dot X_t|$, and the identity $\eta_s = \dot\theta_t - k_t\xi$ gives

$$\|\eta\|_\infty \le CL\varepsilon, \qquad \|\eta_s\|_\infty \le C(1+L)\varepsilon.$$

Since the full perimeter is $2L$, integrating over s and t completes the proof. □

The map $\mathcal{B}$ does not increase W and bounds two arclength derivatives of the rear normal velocity using at most one arclength derivative of the front normal velocity.

Lemma 6.2. *Fix $0 < \widehat{K} < 1$. Let Γ be a smooth path of smooth regular convex closed fronts with curvature at most $\widehat{K}$. Their images under $\mathcal{B}$ form a smooth path, and*

$$W(\mathcal{B}\Gamma) \le W(\Gamma), \tag{6.1}$$
$$S_0(\mathcal{B}\Gamma) \le C_0 W(\Gamma), \tag{6.2}$$
$$S_1(\mathcal{B}\Gamma) \le C_1(W(\Gamma) + S_0(\Gamma)), \tag{6.3}$$
$$S_2(\mathcal{B}\Gamma) \le C_2(W(\Gamma) + S_0(\Gamma) + S_1(\Gamma)). \tag{6.4}$$

The constants depend only on $\widehat{K}$.

Proof. Smooth dependence follows from Lemma 2.1. At a fixed time, let x be rear arclength, write $\dot{R} = \xi\tau + \eta_R\nu$, and denote the front normal velocity by η_F. Since $F = R + \tau$ and $\dot{\tau} = (\eta_{R,x} + k\xi)\nu$, we have

$$\dot{F} = \xi\tau + (\eta_R + \eta_{R,x} + k\xi)\nu, \qquad \nu_F = \cos\delta\,\nu - \sin\delta\,\tau.$$

Since $k = \tan\delta$, taking the inner product of $\dot{F}$ with ν_F cancels the tangential terms and gives

$$(1 + \partial_x)\eta_R = \sec\delta\,(\eta_F \circ s), \tag{6.5}$$

where $s = s(x)$ is the corresponding front arclength. Let ℓ be the rear perimeter. On $\mathbf{R}/\ell\mathbf{Z}$, the inverse of $1 + \partial_x$ is

$$(\mathcal{R}_\ell f)(x) = \frac{1}{1 - e^{-\ell}} \int_{x-\ell}^{x} e^{-(x-t)} f(t)\,dt.$$

Its kernel is positive and has integral 1. Since $dx = \cos\delta\,ds$, we have

$$\int |\sec\delta\,(\eta_F \circ s)|\,dx = \int |\eta_F|\,ds,$$

which proves (6.1).

Lemma 2.1 bounds $\sec\delta$ and the rear curvature in terms of $\widehat{K}$, so the total turning of the rear tangent gives

$$\ell \ge \frac{2\pi\sqrt{1 - \widehat{K}^2}}{\widehat{K}} > 0.$$

Thus $\mathcal{R}_\ell$ maps L^1 to L^∞ with a uniform bound, proving (6.2). Equation (6.5) then gives (6.3), and differentiating (6.5) once yields

$$\eta_{R,xx} = (\sec\delta)_x\,(\eta_F \circ s) + \sec^2\delta\,(\eta_{F,s} \circ s) - \eta_{R,x}.$$

The identity $\delta_x = \sec\delta\,(K - \sin\delta)$ bounds the coefficients using only $\widehat{K}$ and proves (6.4). No bound on the perimeter of the front or on derivatives of its curvature is used. □

Lemma 6.3. *Let G_j be regular closed curves joined by piecewise smooth paths Γ_j. Suppose that all intermediate curvatures are bounded in absolute value by a common constant, and that*

$$\sum_j (S_0(\Gamma_j) + S_1(\Gamma_j) + S_2(\Gamma_j)) < \infty.$$

Then G_j converges in geometric C^2 to a regular closed C^2 curve.

Proof. After choosing parametrizations that agree at the junctions, concatenate the paths. On each segment, remove the tangential velocity by reparametrization. With $g = |X_u|$, the change of parameter solves

$$\partial_t \varphi = -\frac{\xi \circ \varphi}{g \circ \varphi}.$$

Initializing each segment with the endpoint parametrization of the previous segment leaves S_0, S_1, and S_2 unchanged and gives $\partial_t X = \eta\nu$ in a single transported 1-periodic parameter u. The inward-normal variation formulas are

$$\partial_t \tau = \eta_s \nu, \qquad \partial_t k = \eta_{ss} + k^2 \eta, \qquad \partial_t \log g = -k\eta. \tag{6.6}$$

Summability makes X, τ, k, and $\log g$ uniformly Cauchy. The limiting speed is positive because $\log g$ has a finite uniform limit.

Let X_j, g_j, τ_j, ν_j, and k_j be the resulting quantities on G_j. Switch to normalized arclength on $[0,1]$ by setting

$$\ell_j = \int_0^1 g_j(u)\,du, \qquad h_j(u) = \ell_j^{-1} \int_0^u g_j(v)\,dv, \qquad Y_j = X_j \circ h_j^{-1}.$$

The maps h_j converge in C^1 to a map with positive derivative. Consequently,

$$Y_j' = \ell_j \tau_j \circ h_j^{-1}, \qquad Y_j'' = \ell_j^2 (k_j \nu_j) \circ h_j^{-1}$$

converge uniformly, as does Y_j. Their limit is a regular closed C^2 parametrization. □

Theorem 6.4. *Let Q_n be centrally symmetric ovals centered at the origin, and put $A_n = \mathcal{B}Q_{n+1}$. Suppose that Q_n and A_n have the same half-perimeter L_n, and that all these curves have curvature at most a common constant $K_0 < 1$. For each pair, choose arclength origins at which the tangent directions of Q_n and A_n agree, and define*

$$e_n = (1 + L_n)^2 \|k_{Q_n} - k_{A_n}\|_{L^1(0,L_n)}, \qquad r_n = \sum_{m=n}^{\infty} e_m.$$

There are constants $\eta_*, C > 0$, depending only on K_0, with the following property. If $r_0 \le \eta_*$, then all backward iterates

$$Z_n^{(N)} = \mathcal{B}^{N-n} Q_N \qquad (N \ge n)$$

are defined. For each fixed n, the curves $Z_n^{(N)}$ converge in geometric C^2 as $N \to \infty$ to a smooth oval X_n. These limits satisfy

$$X_n = \mathcal{B}X_{n+1}, \qquad \mathcal{T}X_n = X_{n+1},$$

and

$$d_{\mathrm{H}}(X_n, Q_n) \le C r_n, \tag{6.7}$$

where d_{H} denotes Hausdorff distance.

Proof. Interpolation paths. By Lemma 6.1, there are smooth paths Λ_n of centered ovals from Q_n to A_n with curvature at most K_0 and

$$W(\Lambda_n) + S_0(\Lambda_n) + S_1(\Lambda_n) \le d_n, \qquad d_n = C_{\mathrm{int}} e_n.$$

Put $a_n = \sum_{m=n}^{\infty} d_m = C_{\mathrm{int}} r_n$. All constants below depend only on K_0.

Set $\widehat{K} = (1 + K_0)/2$, and take the constants C_0, C_1, and C_2 from Lemma 6.2 for this curvature bound. Put

$$M_0 = \max\{1, C_0\}, \qquad M_1 = \max\{1, C_1(1 + M_0)\}, \qquad C_* = C_2(1 + M_0 + M_1) + \widehat{K}^2\, C_0,$$

and choose $\eta_* > 0$ so small that $C_*\, C_{\mathrm{int}}\, \eta_* < \widehat{K} - K_0$.

Invariant family. Let $\mathcal{D}_n$ consist of the curves joined to Q_n by a piecewise smooth path of centered ovals with curvature at most $\widehat{K}$ and

$$W \le a_n, \qquad S_0 \le M_0 a_n, \qquad S_1 \le M_1 a_n.$$

We claim that

$$\mathcal{B}(\mathcal{D}_{n+1}) \subseteq \mathcal{D}_n. \tag{6.8}$$

Let $G \in \mathcal{D}_{n+1}$, and let Γ be a path from Q_{n+1} to G satisfying the definition of $\mathcal{D}_{n+1}$. By Lemma 2.1, the path $\mathcal{B}\Gamma$ is defined and starts at A_n. Lemma 6.2 gives

$$\begin{aligned} W(\mathcal{B}\Gamma) &\le a_{n+1}, & S_0(\mathcal{B}\Gamma) &\le C_0 a_{n+1}, \\ S_1(\mathcal{B}\Gamma) &\le C_1(1 + M_0) a_{n+1}, & S_2(\mathcal{B}\Gamma) + \widehat{K}^2\, S_0(\mathcal{B}\Gamma) &\le C_* a_{n+1}. \end{aligned}$$

The path $\mathcal{B}\Gamma$ starts at A_n with curvature at most K_0. As long as its curvature remains at most $\widehat{K}$, equation (6.6) in a parametrization with zero tangential velocity gives

$$\|k(t) - k(0)\|_\infty \le S_2(\mathcal{B}\Gamma) + \widehat{K}^2\, S_0(\mathcal{B}\Gamma) \le C_* a_{n+1} < \widehat{K} - K_0,$$

so the curvature stays below $\widehat{K}$ throughout $\mathcal{B}\Gamma$. Since $a_n = d_n + a_{n+1}$, concatenating Λ_n with $\mathcal{B}\Gamma$ gives a path that satisfies the bounds defining $\mathcal{D}_n$, proving (6.8). Hence every $Z_n^{(N)}$ is defined and belongs to $\mathcal{D}_n$.

Convergence. Every curve on Λ_N belongs to $\mathcal{D}_N$, because the path up to that curve satisfies the defining bounds with $d_N \le a_N$. Repeated use of (6.8) therefore shows that

$$\Gamma_{n,N} = \mathcal{B}^{N-n} \Lambda_N$$

exists, stays in $\mathcal{D}_n$, and joins $Z_n^{(N)}$ to $Z_n^{(N+1)}$. Iterating Lemma 6.2 gives

$$W(\Gamma_{n,N}) \le d_N, \qquad S_0(\Gamma_{n,N}) \le M_0 d_N, \qquad S_1(\Gamma_{n,N}) \le M_1 d_N.$$

For $N \ge n + 1$, we have $\Gamma_{n,N} = \mathcal{B}\Gamma_{n+1,N}$, so (6.4) gives

$$S_2(\Gamma_{n,N}) \le C_2(1 + M_0 + M_1) d_N.$$

For fixed n, Lemma 6.3 therefore applies to the sequence with $N \ge n + 1$. Let X_n be its geometric C^2 limit.

The limit X_n has curvature $0 \le k \le \widehat{K} < 1$, and its unit tangent has turning number 1 because the unit tangent maps converge uniformly. Relative to the tangent at each point of X_n, the tangent angle increases from 0 to 2π around the curve, so projection onto the inward normal first increases from 0 and then decreases back to 0. Every tangent line is therefore supporting, and X_n bounds a convex body.

Since $\mathcal{B}$ is continuous in geometric C^2 by Lemma 2.1, the identities $Z_n^{(N)} = \mathcal{B} Z_{n+1}^{(N)}$ pass to the limit to give

$$X_n = \mathcal{B} X_{n+1}, \qquad \mathcal{T} X_n = X_{n+1}.$$

By Lemma 2.1 again, each X_n has positive curvature and gains one derivative at each step $X_n = \mathcal{B}X_{n+1}$, so iterating $X_n = \mathcal{B}^r X_{n+r}$ shows that every X_n is smooth.

Distance from the comparison curves. Every $Z_n^{(N)}$ can be joined to Q_n by a path in $\mathcal{D}_n$. In a parametrization with zero tangential velocity, each point moves a distance at most $S_0 \le M_0 a_n$, so $d_{\mathrm{H}}(Z_n^{(N)}, Q_n) \le M_0 a_n$. Passing to the limit and using $a_n = C_{\mathrm{int}} r_n$ proves (6.7). □

If $r_0 < \infty$, then $r_n \to 0$, so the theorem applies after discarding finitely many initial indices.

7 Proof of the main theorem

Proof of Theorem 1.1. Use the fixed translating hairpin from Theorem 3.4. By Proposition 4.3, the half-perimeter function P is continuous and satisfies

$$\frac{\Delta}{2} \le H - P(H) \le \frac{3\Delta}{2}$$

for all sufficiently large H. By the intermediate value theorem, each sufficiently large H_0 determines a sequence $(H_n)_{n\ge 0}$ with

$$P(H_{n+1}) = H_n, \qquad H_n \ge H_0 + \frac{\Delta}{2} n. \tag{7.1}$$

Translate each pair (R_{H_n}, F_{H_n}) so that its center of symmetry is the origin, and set $Q_0 = F_{H_0}$. Inductively rotate the pair $(R_{H_{n+1}}, F_{H_{n+1}})$ so that the rear tangent and the tangent of Q_n agree at $u = 0$ in the arclength coordinates used in Theorem 5.1. Define

$$Q_{n+1} = F_{H_{n+1}}, \qquad A_n = \mathcal{B}Q_{n+1} = R_{H_{n+1}}.$$

Because $\mathcal{T}$ commutes with translations and rotations, these choices preserve the relations $\mathcal{T}R_{H_n} = F_{H_n}$ and all curvature estimates. By (7.1), Q_n and A_n have the same half-perimeter H_n, and their chosen tangent directions agree.

Theorem 5.1 gives

$$e_n \le C(1 + H_n)^2 e^{-\beta H_{n+1}}.$$

By (7.1), decreasing β to absorb the polynomial factor yields, for some $\beta' > 0$,

$$r_0 \le C \sum_{n=0}^{\infty} e^{-\beta'(H_0 + n\Delta/2)} \le C' e^{-\beta' H_0} \longrightarrow 0 \qquad (H_0 \to \infty).$$

All curves Q_n and A_n have curvature at most $K_0 = 1/10$. For sufficiently large H_0, Theorem 6.4 gives smooth ovals X_n with $\mathcal{T}X_n = X_{n+1}$. Hausdorff distance d changes the width in each direction by at most $2d$, so Lemma 4.4 and (6.7) show that each X_n has width at most $C_W + 2Cr_n \le C_W + 2Cr_0$ in some direction. Since this bound is independent of n, whereas the nth iterate of a circle of radius r is a circle of radius $\sqrt{r^2 + n}$ and has width $2\sqrt{r^2 + n}$ in every direction, $\Gamma_0 = X_0$ is not a circle. □

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